Computer Networks (CSE3006) - Tutorial 1: Solutions

Programme	B.Tech. [CSE]	**Semester**	Fall Semester 2026-2027
Course	Computer Networks	**Code**	CSE3006
Faculty	Dr. S. Periyanayagi	**Tutorial No.**	1

Q1. Efficiency of a 5-layer system (100-byte message, 10-byte header per layer)

Total bytes transmitted = data + header added at each of the 5 layers:

Total = 100 + 5(10) = 150 bytes

Efficiency = application bytes / total bytes transmitted = 100 / 150

Answer: Efficiency = 0.667 = 66.7%

Q2. Bit rate for digitized voice channel (4 kHz BW, Nyquist sampling, 8 bits/sample)

Sampling rate (Nyquist) = 2 x f_max = 2 x 4 kHz = 8000 samples/sec

Bit rate = sampling rate x bits/sample = 8000 x 8

Answer: Bit rate = 64,000 bps = 64 kbps

Q3. AM modulation: audio signal 5 kHz, carrier 500 kHz

a. USB = f_c + f_m = 500 + 5 = **505 kHz**

b. LSB = f_c - f_m = 500 - 5 = **495 kHz**

c. Bandwidth = 2 x f_m = 2 x 5 kHz

Answer: USB = 505 kHz, LSB = 495 kHz, Bandwidth = 10 kHz

Q4. Baud rate at bit rate 9600 bps

Baud rate S = N / n, where N = bit rate and n = bits per symbol

a. BPSK (n = 1): S = 9600 / 1 = 9600 baud

b. QPSK (n = 2): S = 9600 / 2 = 4800 baud

c. 8-PSK (n = 3): S = 9600 / 3 = 3200 baud

d. Higher bits/symbol -> lower baud rate needed for the same bit rate, so 8-PSK uses the least bandwidth.

Answer: BPSK = 9600 baud, QPSK = 4800 baud, 8-PSK = 3200 baud; 8-PSK has the highest bandwidth efficiency.

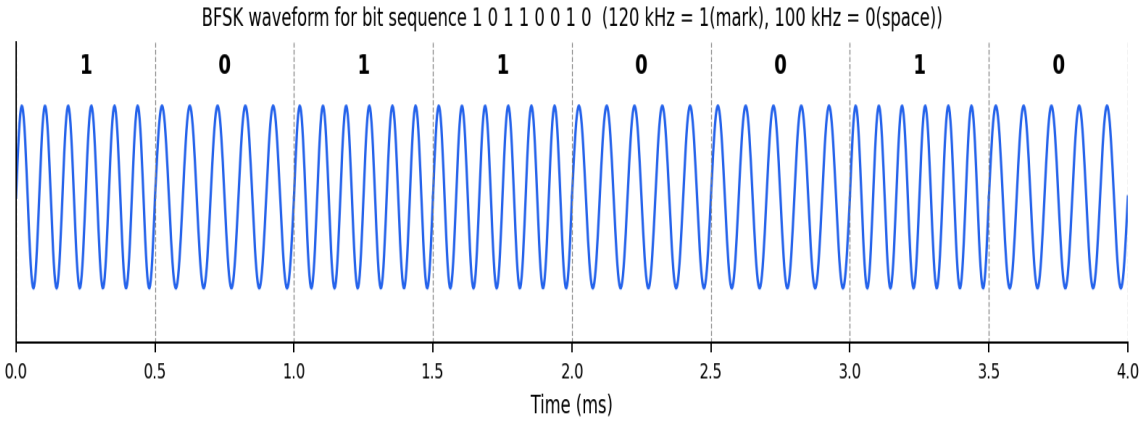
Q5. Frequency of the signal (period = 4 ms shown in figure)

Frequency f = 1 / T = 1 / (4 x 10^-3 s)

Answer: f = 250 Hz

Q6. Binary FSK: mark = 120 kHz (bit 1), space = 100 kHz (bit 0), T_b = 0.5 ms, sequence 10110010

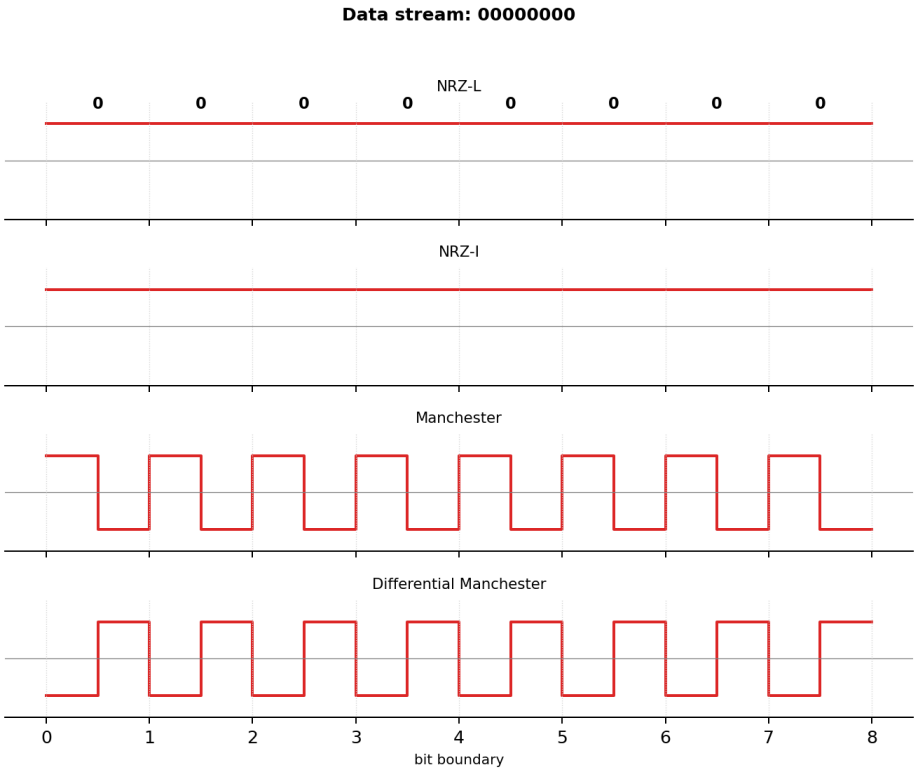
Bit	1	0	1	1	0	0	1	0
Freq. transmitted	120 kHz	100 kHz	120 kHz	120 kHz	100 kHz	100 kHz	120 kHz	100 kHz



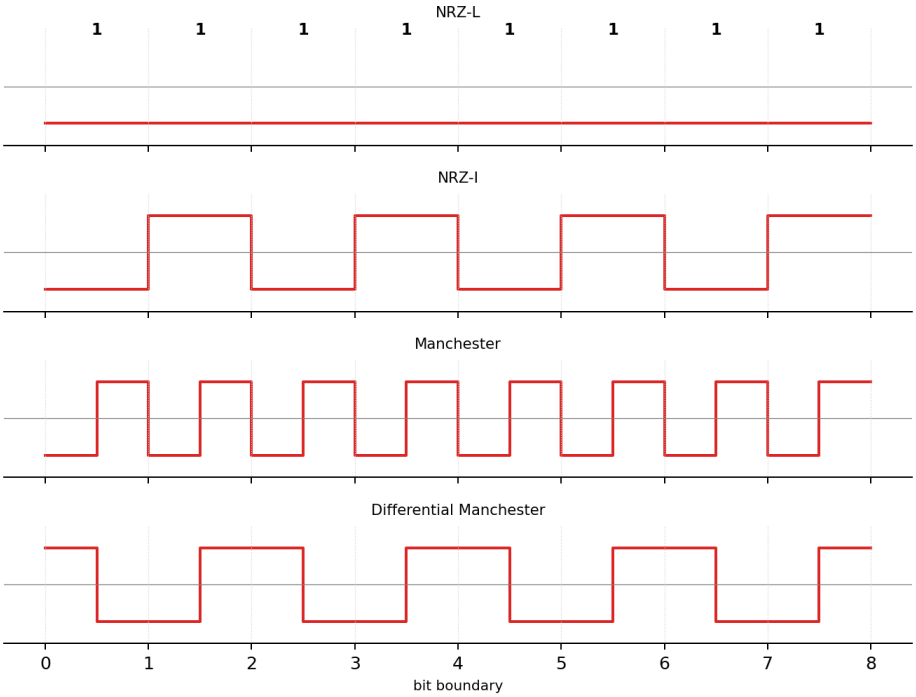
b. FSK waveform (frequency shifts between 120 kHz and 100 kHz each 0.5 ms bit interval)

Q7. NRZ-L, NRZ-I, Manchester and Differential Manchester encoding

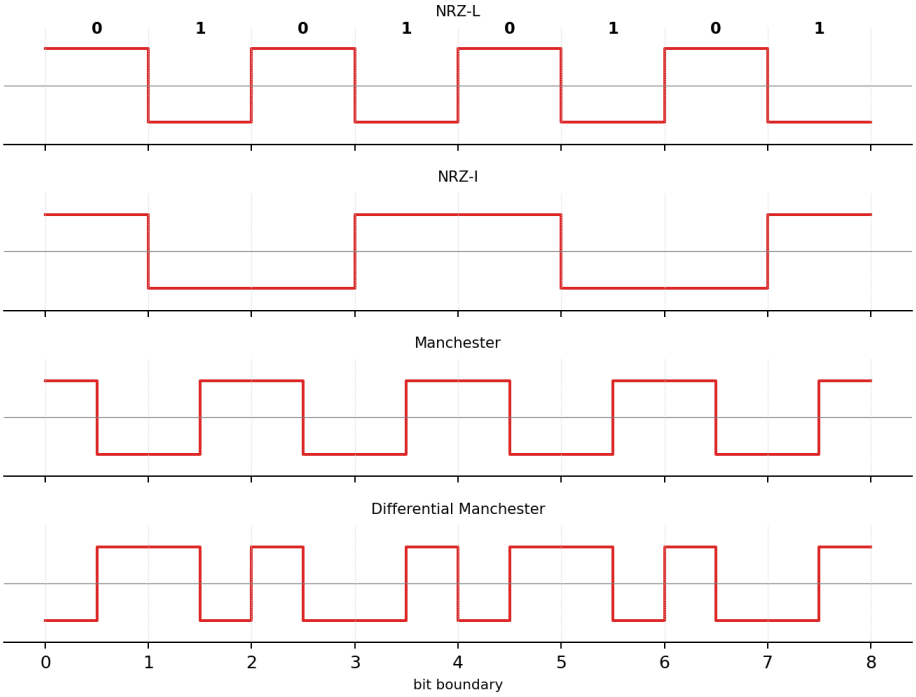
Conventions used: **NRZ-L**: 0 = high, 1 = low. **NRZ-I**: 1 = transition (level inverts), 0 = no change. **Manchester**: mid-bit transition, 1 = low-to-high, 0 = high-to-low. **Differential Manchester**: always a mid-bit transition; 0 = extra transition at start of bit, 1 = no transition at start.

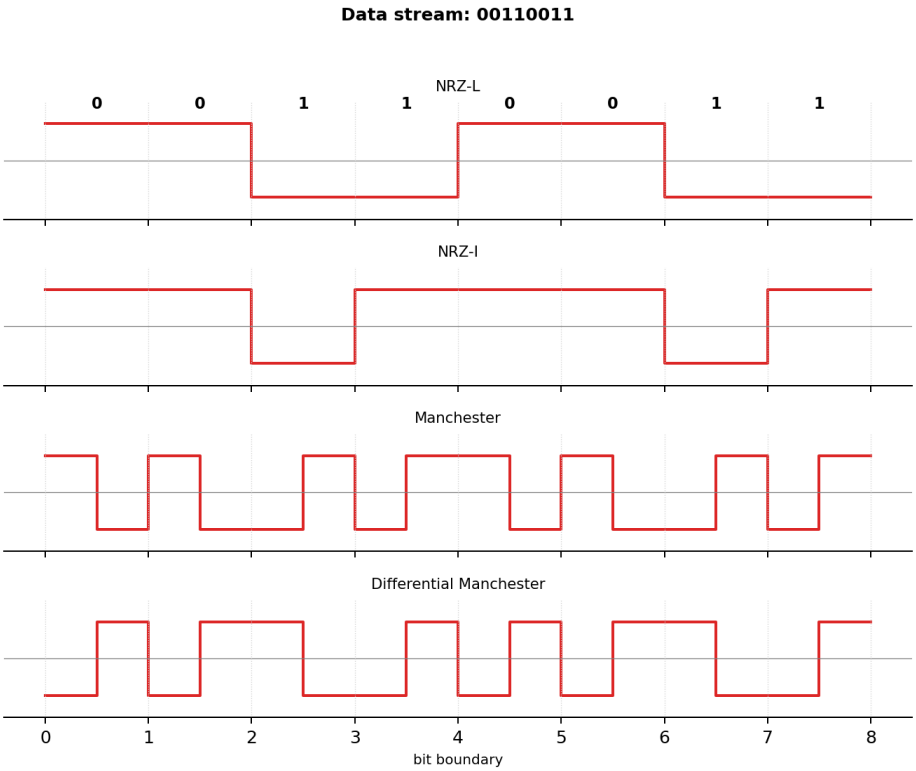


Data stream: 11111111



Data stream: 01010101





Q8. Low-pass signal, BW = 200 kHz, 1024 quantization levels

Nyquist sampling rate = 2 x 200 kHz = 400,000 samples/sec

Bits per sample n = log2(1024) = 10 bits

a. Bit rate = 400,000 x 10 = 4,000,000 bps = **4 Mbps**

b. SNR(dB) = 6.02n + 1.76 = 6.02(10) + 1.76 = 60.2 + 1.76 = **61.96 dB approx. 62 dB**

c. Minimum PCM (baseband) bandwidth = bit rate / 2 = 4,000,000 / 2 = **2,000,000 Hz = 2 MHz**

Answer: Bit rate = 4 Mbps, SNRdB approx. 62 dB, PCM bandwidth = 2 MHz

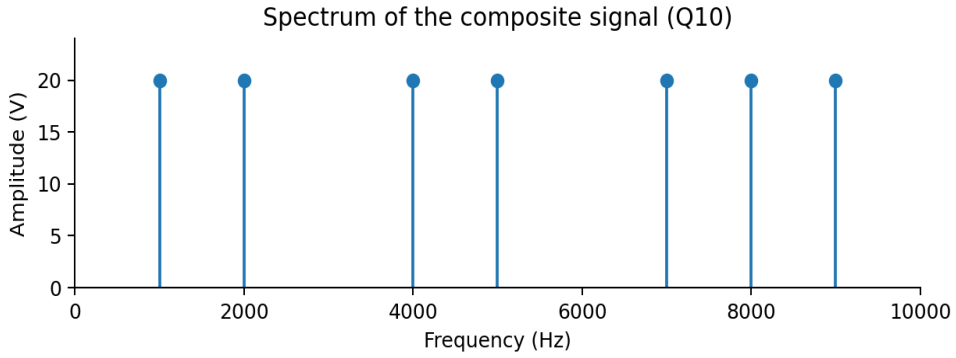
Q9. Baud rate S = N/n for the given bit rate N and modulation

	Bit rate (N)	Scheme	bits/symbol (n)	Baud rate S = N/n
a	2000 bps	FSK	1	2000 baud
b	4000 bps	ASK	1	4000 baud
c	6000 bps	QPSK	2	3000 baud
d	36,000 bps	64-QAM	6	6000 baud

Q10. Bandwidth of a periodic signal with components 1,2,4,5,7,8,9 kHz

Bandwidth = f_max - f_min = 9000 - 1000

Answer: Bandwidth = 8000 Hz = 8 kHz

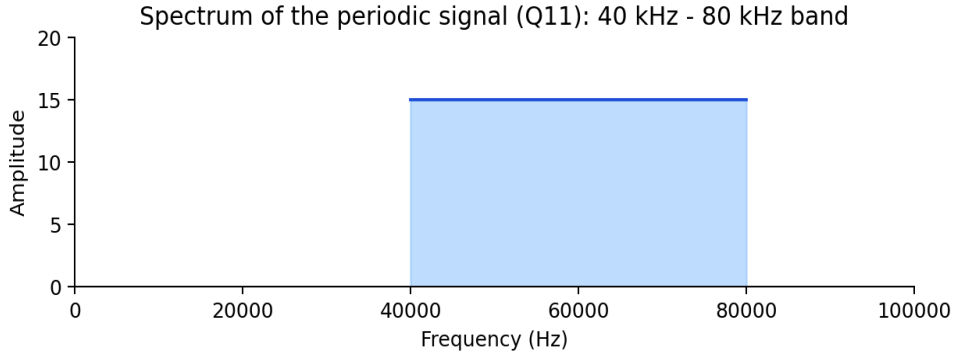


Spectrum: 7 discrete lines, each of amplitude 20 V

Q11. Periodic signal: bandwidth = 40 kHz, highest frequency = 80 kHz

Lowest frequency = highest frequency - bandwidth = 80 - 40

Answer: Lowest frequency = 40 kHz (signal spans 40 kHz to 80 kHz)



Spectrum: continuous band from 40 kHz to 80 kHz, equal amplitude

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Q12. Propagation and transmission time (5 MB message, 1 Mbps link, 12,000 km, v = 2.4x10^8 m/s)

Message size = 5 MB = 5 x 8 x 10^6 bits = 40 x 10^6 bits

Transmission time = data size / bandwidth = 40 x 10^6 / 1 x 10^6

Propagation time = distance / speed = (12,000 x 10^3) / (2.4 x 10^8)

Answer: Transmission time = 40 s; Propagation time = 0.05 s = 50 ms

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Q13. 64-QAM system transmitting at 48 Mbps

a. Bits per symbol n = log2(64) = **6 bits/symbol**

b. Baud rate = bit rate / n = 48 x 10^6 / 6 = **8 x 10^6 = 8 Mbaud**

c. For 256-QAM: n = log2(256) = 8 bits/symbol. Baud rate = 48 x 10^6 / 8 = **6 x 10^6 = 6 Mbaud**

d. 256-QAM carries more bits per symbol, so it needs a lower baud rate (less bandwidth) for the same bit rate -> 256-QAM is more bandwidth efficient.

Answer: 6 bits/symbol; 8 Mbaud for 64-QAM; 6 Mbaud for 256-QAM; 256-QAM is more bandwidth efficient.

Q14. Nyquist sampling rate

a. Low-pass signal, BW = 200 kHz (f_min = 0, f_max = 200 kHz): Nyquist rate = 2 x f_max = 2 x 200 kHz = **400 kHz**

b. Band-pass signal, BW = 200 kHz, f_min = 100 kHz -> f_max = f_min + BW = 100 + 200 = 300 kHz. Nyquist rate = 2 x f_max = 2 x 300 kHz = **600 kHz**

Answer: Low-pass: 400 kHz; Band-pass: 600 kHz

Q15. Manchester encoding at 100 kbps

a. Manchester needs 2 signal transitions per bit on average -> minimum signal (baud) rate = 2 x bit rate = 2 x 100 kbps = **200,000 baud**

b. Manchester is a biphase code, so its minimum required bandwidth equals the baud rate: B(min) = **200 kHz**

c. Polar NRZ-L: baud rate = bit rate = 100,000 baud; minimum bandwidth = baud rate / 2 = 100,000/2 = **50 kHz**

d. Polar NRZ-L needs less bandwidth (50 kHz vs 200 kHz), so it is more bandwidth efficient. However, Manchester provides built-in clock synchronization (self-clocking) and has no DC component, which NRZ-L lacks.

Answer: Manchester: 200 kbaud, 200 kHz BW. NRZ-L: 100 kbaud, 50 kHz BW. NRZ-L is more bandwidth efficient; Manchester is better for synchronization.

End of solutions.